The marked two-swap saturation cases

Human Compression project: bounded L044 proof component

Use the accepted block-deficit setting: $G \le A_n$ is a transitive string C-group of degree $n \ge 12$, every maximal omission is intransitive, and there is a unique eligible i-edge with endpoints a, b in the two G_i-orbits A, B. Suppose both orbits have two blocks, of sizes $k, l \ge 2$, and $\alpha = \beta = 0$. Put $K = k+l$ and $s = \rho_i$. The accepted saturation lemma gives even k, l and a faithful $M \cong C_2^2$, acting on each orbit as the diagonal two-block image. We preserve the literal distinguished generators; no product replacement is made.

Lemma 1 (The marked basis is pure)**.** *After interchanging A, B if necessary, $M = \{\tau_A, \tau_B\}$ with $\tau_A = \rho_{i-1}$ exchanging the blocks of A and fixing B pointwise, and $\tau_B = \rho_{i+1}$ doing the reverse. Both $x = \rho_{i-2}$ and $y = \rho_{i+2}$ exist.*

Proof. Every generator outside $i-1, i+1$ fixes both crossing endpoints, so its images on the two two-block systems are trivial. The two independent images in M must therefore occur at these two neighboring indices. They commute.

Suppose both neighboring generators exchange the blocks of A. Since $\beta = 0$, they act identically there: each swaps the two points in every row. Let a' be the point opposite a. Every other generator except possibly x, y commutes with both swaps, and so fixes a' as well as a. The generator x, if present, commutes with ρ_{i+1}, and y, if present, commutes with ρ_{i-1}; each also fixes a'. Thus all of G_i preserve $\{a, a'\}$, contradicting transitivity on A because $k \ge 2$. The same argument applies to B. Each block character is therefore nontrivial on exactly one of the two generators; independence of their simultaneous images makes those choices different. This is precisely the asserted pure basis in the original ordering.

Every generator other than x, τ_A fixes a': for y and all distant generators use commutation with τ_A, while τ_B is trivial on A. If x were absent, the pair $\{a, a'\}$ would again be invariant. Hence x exists, and must move a' to another row. Symmetrically y exists and moves the opposite point b' to another row of B. □

Lemma 2 (Two-row orbits are excluded)**.** *In this setting neither k nor l can equal two.*

Proof. Suppose $k = 2$. The generator x fixes a and preserves each of the two blocks of A, so is the identity on the first block. Since it must move a', it is exactly the single transposition on the other block. On B, the generator x commutes with τ_B and preserves the two blocks, so acts diagonally and has even sign. Thus x is odd on all points, a contradiction. The proof for $l = 2$ is symmetric. □

Let C be the generators outside x, τ_A, s, τ_B, y. They are diagonal on the row pairs of both original orbits, and fix both row anchors. Their row action is faithful, so $\langle C \rangle \le S_{k-1} \times S_{l-1}$, with its full independence and IP retained.

Lemma 3 (The fully saturated rank is impossible)**.** *One cannot have $\gamma = 0$, or equivalently $r = K + 1$.*

Proof. By the preceding lemma take $k, l \ge 4$. With $\gamma = 0$, the family C has $K - 4$ generators, the maximal independent length in $S_{k-1} \times S_{l-1}$. The accepted equality clause identifies its group as that full product. Its prefix P at indices at most $i-3$ and suffix Q at indices at least $i+3$

commute and have trivial intersection by IP. Since the two symmetric factors are centerless and directly indecomposable, the interval groups are products of the whole factors.

The first row factor must occur in P. Otherwise x centralizes its full natural S_{k-1}, whose action on each k-point block has a unique fixed point and trivial centralizer. This would make x fix the opposite anchor a', a contradiction. Likewise the second factor must occur in Q. Thus $P = S_{k-1}$ and $Q = S_{l-1}$, each with a maximal independent string marking.

The generator x fixes B pointwise: its row action there centralizes the natural Q and fixes its unique row anchor. If $k \geq 6$, the ordinary Coxeter marking of S_{k-1} has far-end omission S_{k-2} fixing the anchor and one other row. There is no outer degree-six exception because k is even. Since x centralizes that omission and fixes a, it is the identity on the first block of A and the single transposition involving a' on the other. It is odd, impossible. This uses the accepted L047 maximal-string classification, including degree five. Symmetrically $l \geq 6$ is impossible.

The only case left is $k = l = 4$. On A, the prefix has generators $(1\ 2)(1'\ 2')$ and $(2\ 3)(2'\ 3')$ on the three nonanchor rows. The generator x centralizes the first and fixes $0 = a$, while it must move $0' = a'$. Its evenness and the fact that it fixes B give precisely

$$(1\ 2)(0'\ 3') \quad \text{or} \quad (1'\ 2')(0'\ 3').$$

As proved explicitly in the accepted four-row endpoint, these three generators have natural symmetric images on the three first-block nonanchor points and all four second-block points, and their centralizer fixes those seven points individually. The generator s commutes with them, so fixes $A \setminus \{a\}$ pointwise. The symmetric argument with Q, y shows that it fixes $B \setminus \{b\}$ pointwise. Consequently $s = (a\ b)$ is odd, the final contradiction. □

Remaining scope. This component proves the pure-basis reduction for $\beta = 0$ without changing the marking and excludes $\gamma = 0$. The case $\beta = 0$, $\gamma = 1$ leaves a faithful independent string of length $K - 5$ in $S_{k-1} \times S_{l-1}$, with $k, l \geq 4$ even; its classification is not asserted here. The case $\beta = 1$ has a regular four-point M-orbit and is also separate. All other outstanding L044 obligations remain pending. This is not a whole-source recovery or a new question claim.

Attribution and paid dependencies. The source problem and orbit-cut framework are from Cameron, Fernandes, Leemans and Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 5.1. The present marked-basis and endpoint arguments are new proofs of bounded cases in that framework. They use the complete accepted block-deficit, small-endpoint and two-row notes, and the full retained L047 extremal/normal-quotient interfaces, including Whiston's independent-generation bounds. Their exact files and accepted reviews are bound in the companion dependency map; no sibling proof is an uncharged premise. No new finite census or formal theorem is used.